EXPERIMENT: Write the prolog program to implement family tree.

Aim

To write and implement a Prolog program for representing and querying a family tree using facts and rules.

Objectives

- To represent family relationships using Prolog facts.
- To define relationships such as parent, father, mother, sibling, brother, sister, grandfather, grandmother, and grandparent.
- To execute queries and retrieve family relationships.
- To understand logical inference in Prolog.

Algorithm

1. Start the program.
2. Define facts for male and female family members.
3. Define parent relationships.
4. Define rules for father and mother.
5. Define rules for siblings, brothers, and sisters.
6. Define rules for grandparents.
7. Load the program into Prolog.
8. Execute queries to find relationships.
9. Display the results.
10. Stop the program.

CODE

```
parents = {
   "john": ["robert", "linda"],
   "mary": ["robert", "linda"],
   "robert": ["mike", "anna"],
   "susan": ["mike", "anna"]
```

```
}
male = ["john", "robert", "mike"]
female = ["mary", "linda", "susan", "anna"]
def is_parent(parent, child):
    if parent in parents and child in parents[parent]:
        return True
    return False
def get_children(parent):
    return parents.get(parent, [])
def get_siblings(person):
    siblings = set()
    for parent, children in parents.items():
        if person in children:
            for child in children:
                if child != person:
                    siblings.add(child)

    return list(siblings)
def get_grandparents(person):
    grandparents = set()
    for parent, children in parents.items():
        if person in children:
            for grandparent, grandchildren in parents.items():
                if parent in grandchildren:
                    grandparents.add(grandparent)
```

```
    return list(grandparents)
print("FAMILY TREE")
print("----------------")

print("John's children:", get_children("john"))
print("Mary's children:", get_children("mary"))
print("Robert's children:", get_children("robert"))
print("Susan's children:", get_children("susan"))

print("\nRelationships")
print("----------------")

print("Is John parent of Robert?", is_parent("john", "robert"))
print("Robert's siblings:", get_siblings("robert"))
print("Mike's grandparents:", get_grandparents("mike"))
print("Anna's grandparents:", get_grandparents("anna"))
```

OUTPUT

```
IDLE Shell 3.12.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug  6 2024, 20:45:27) [MSC v.1940 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\x
dc.py
FAMILY TREE
----------------
John's children: ['robert', 'linda']
Mary's children: ['robert', 'linda']
Robert's children: ['mike', 'anna']
Susan's children: ['mike', 'anna']

Relationships
----------------
Is John parent of Robert? True
Robert's siblings: ['linda']
Mike's grandparents: ['mary', 'john']
Anna's grandparents: ['mary', 'john']
>>> |
                                                                Ln: 18  Col: 0
```

Result

The family tree was successfully implemented in Prolog using facts and rules. The program can determine relationships such as father, mother, sibling, brother, sister, grandfather, grandmother, and child.

Conclusion

The program demonstrates how Prolog facts and rules can represent real-world family relationships. By entering different queries, Prolog uses logical inference to identify relationships between family members.